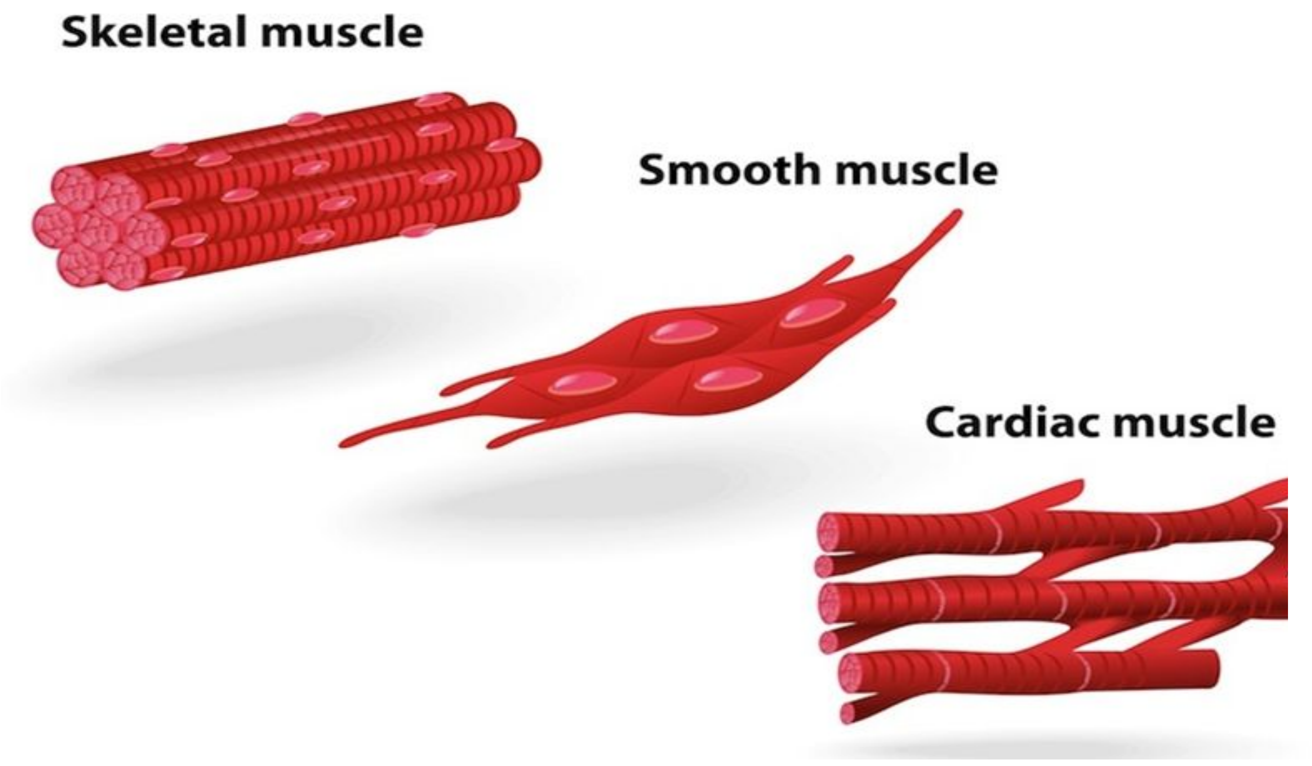
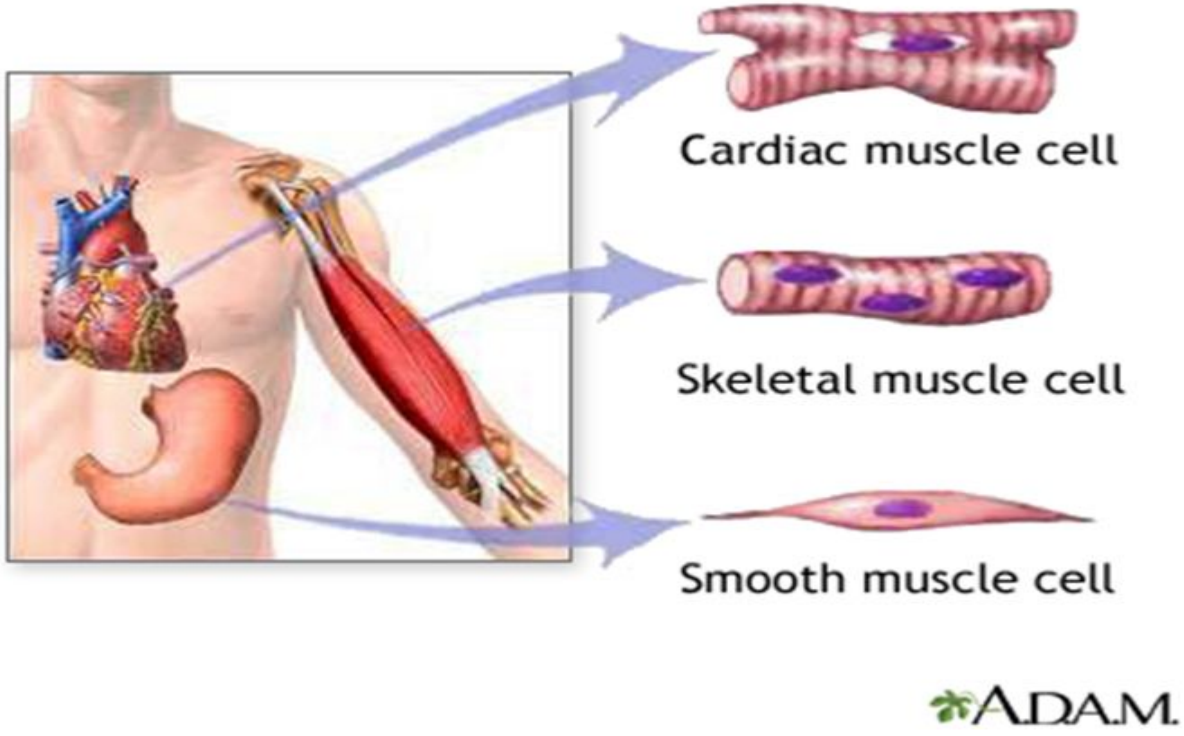
Comparison of skeletal, Smooth and cardiac muscle





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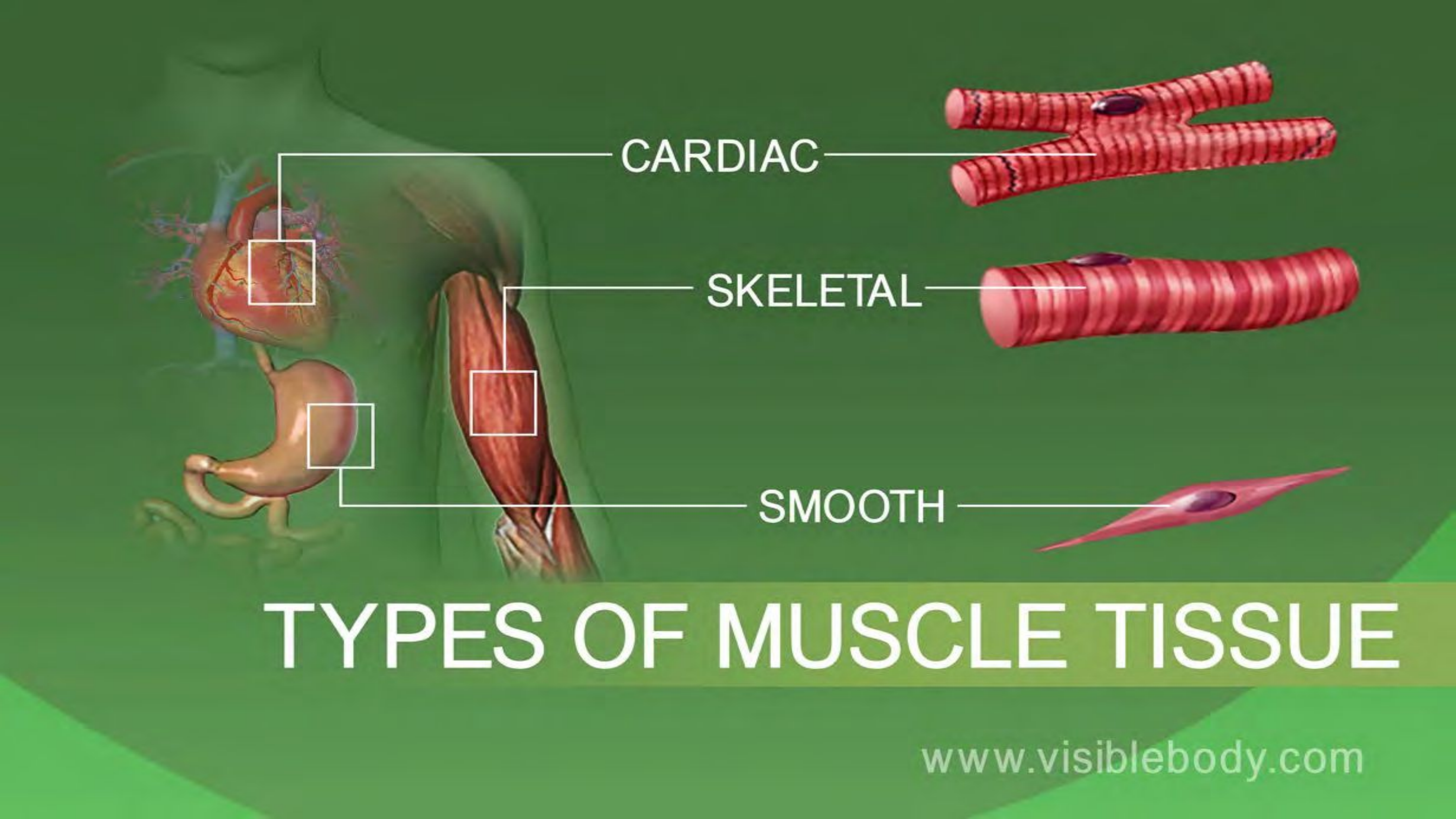
Introduction

- All activities that involve movement depend on muscles
- 650 muscles in the human body
- **Various purposes for muscles for:**
- Voluntary movements: Voluntary movements are actions you control.
- Involuntary movements: Involuntary movements happen automatically without you thinking about them
- Moving, sitting still and standing up straight.
- Digesting food and getting rid of waste (peeing and pooping).
- Pumping blood through your heart and blood vessels.
- Giving birth.
- Muscles also store and release energy your body uses as part of your metabolism.
- Locomotion
- Movement of materials along internal tubes
- Vision and Hearing

The muscular system creates body heat

and also moves the:

- ❖ Bones of the Skeletal system
- ❖ Food through Digestive system
- ❖ Blood through the Circulatory system
- ❖ Fluids through the Excretory system


CARDIAC
SKELETAL
SMOOTH
TYPES OF MUSCLE TISSUE
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- **Skeletal muscle**:

Is in our biceps, triceps, postural muscles, etc.

- **Smooth muscle:**

Is found along our digestive tract: used to move food along

- **Cardiac muscle**:

Is found in the heart

Skeletal muscles

Skeletal muscles are part of your musculoskeletal system

40% of human body mass is made of skeletal muscles.

Skeletal muscles are mainly responsible for locomotion, and voluntary contraction and relaxation

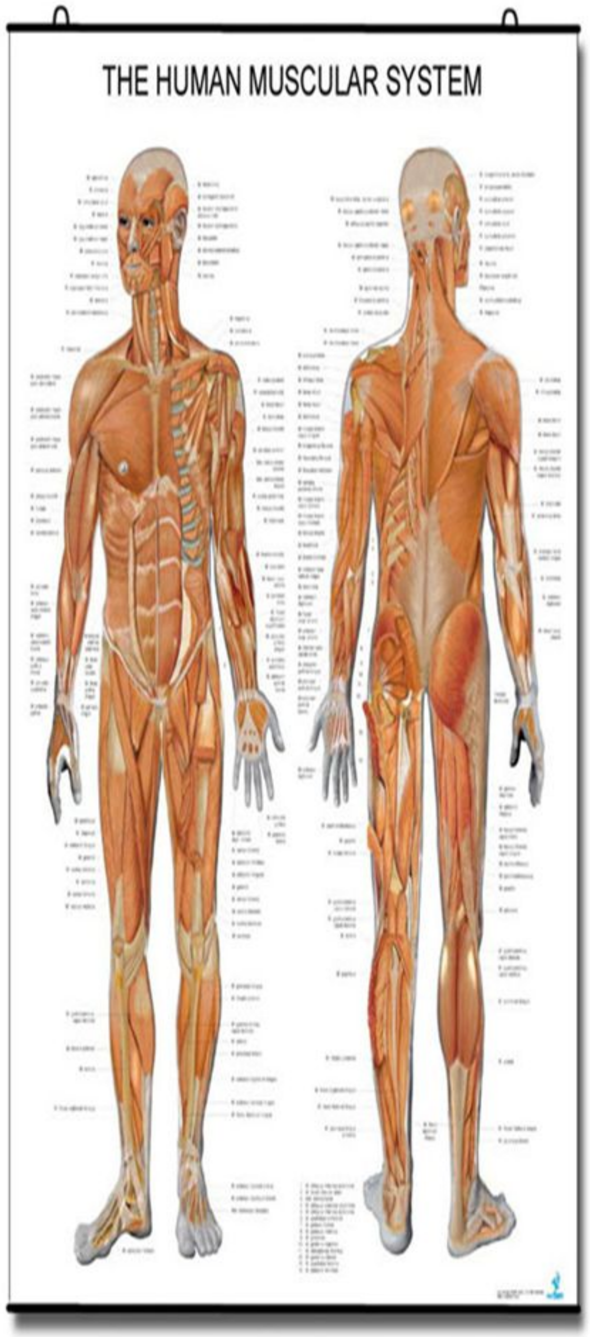
They work with your bones , tendons and ligaments support your weight and move you.

Tendons attach skeletal muscles to bones all over your body.

Skeletal muscles are voluntary—they move when you think about moving that part of your body.

Some muscle fibers contract quickly and use short bursts of energy (fast-twitch muscles).Examples include the gastrocnemius (outer calf) for jumping, quadriceps in sprinters, biceps and pectorals for lifting, and eye muscles for rapid, involuntary movements.

Others move slowly, like your back muscles that help with posture.



Structure of the skeletal muscle

Skeletal muscle is a voluntary, striated muscle tissue attached to bones by tendons, comprising about 35–40% of human body weight.

It powers locomotion, maintains posture, and stabilizes joints through contraction.

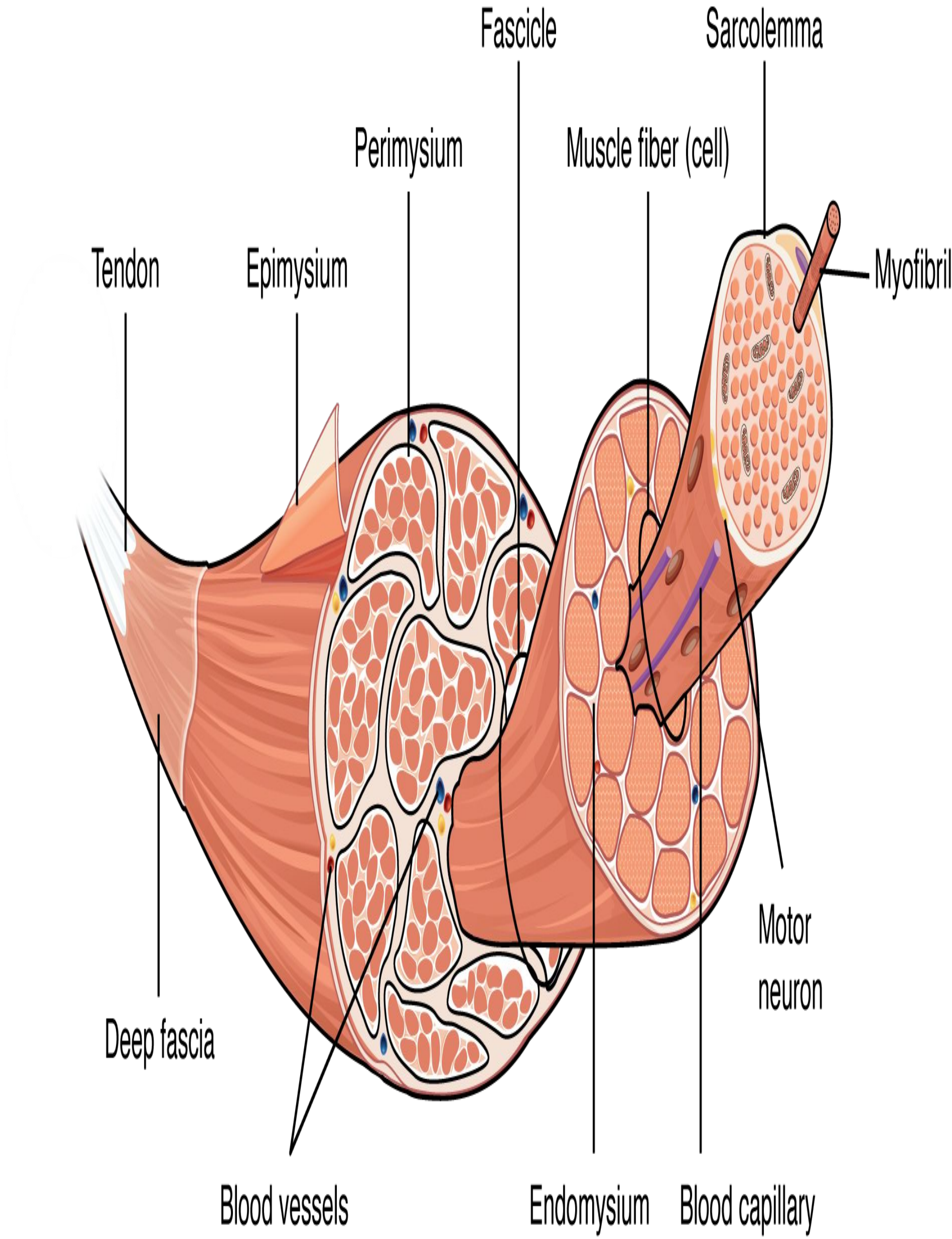
Composed of long, multinucleated fibers (myocytes) organized in bundles (fascicles), these muscles receive neural input to generate movement.

Structure:

Composed of fascicles (bundles of fibers), which contain myofibrils, which are made of contractile units called sarcomeres (thick and thin filaments: myosin and actin).

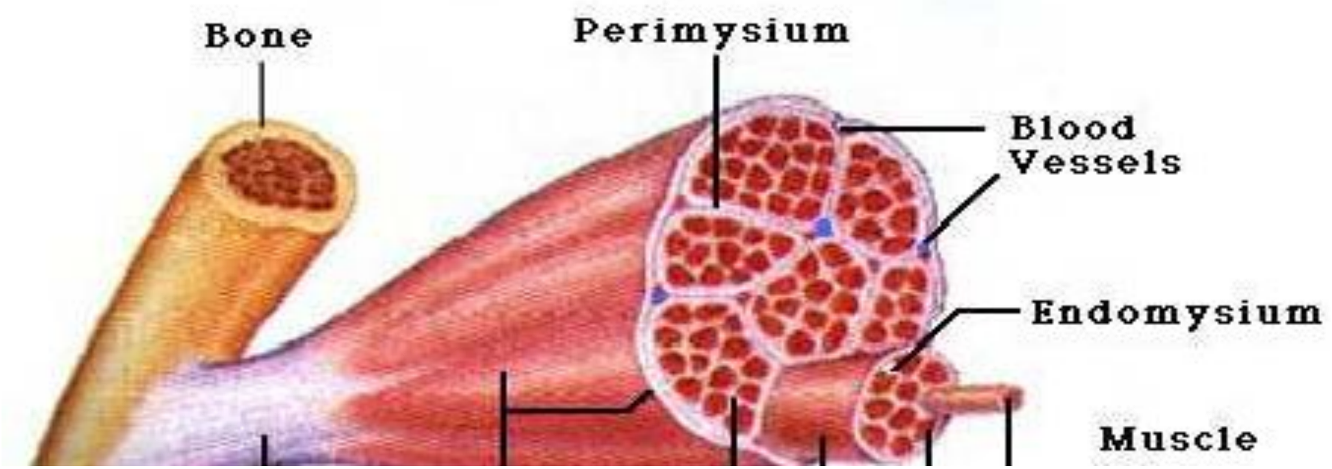
Sarcomeres have alternating thick filaments of myosin that give a dark band (red) and a thin filament of actin that gives a light band (white).

This alternating dark and light band gives rise to the striated appearance of skeletal muscles.

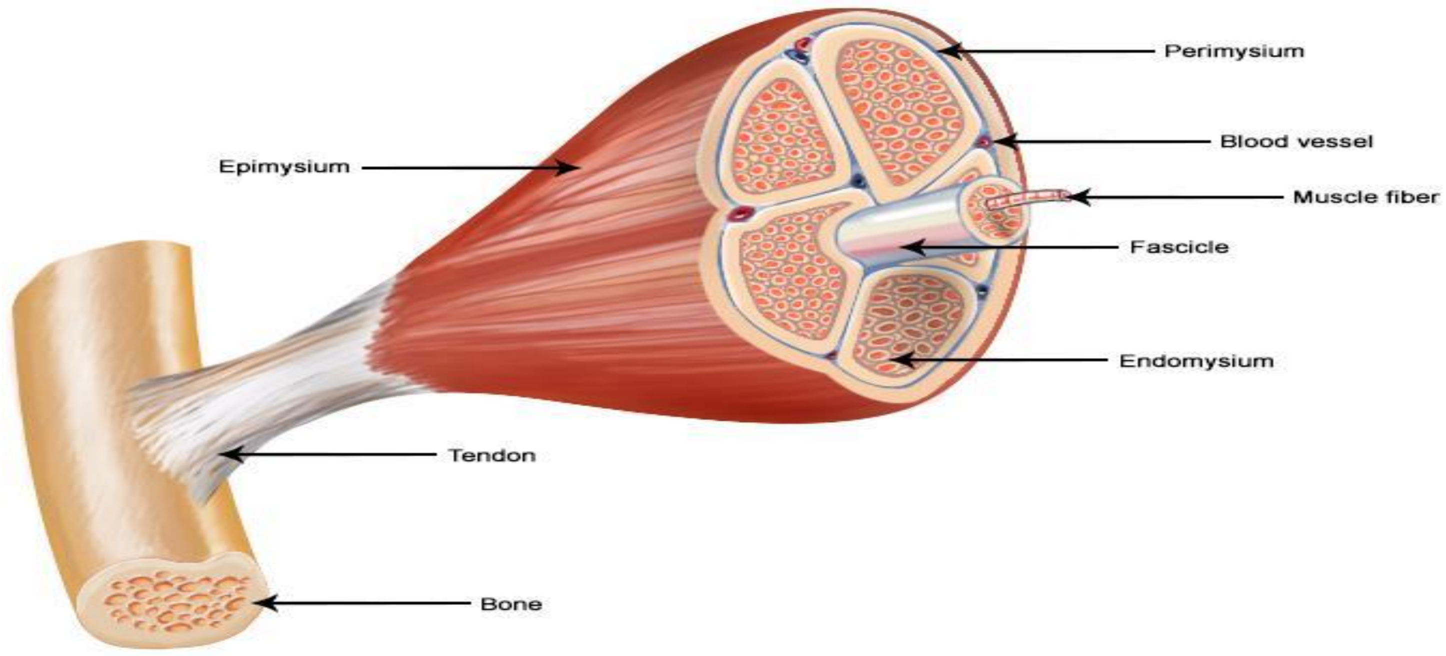


Structure Of The Skeletal Muscle

- Epimysium
  - Surrounds entire muscle
- Perimysium
  - Surrounds bundles of muscle fibers
- Endomysium
  - Surrounds individual muscle fibers



Structure of a Skeletal Muscle



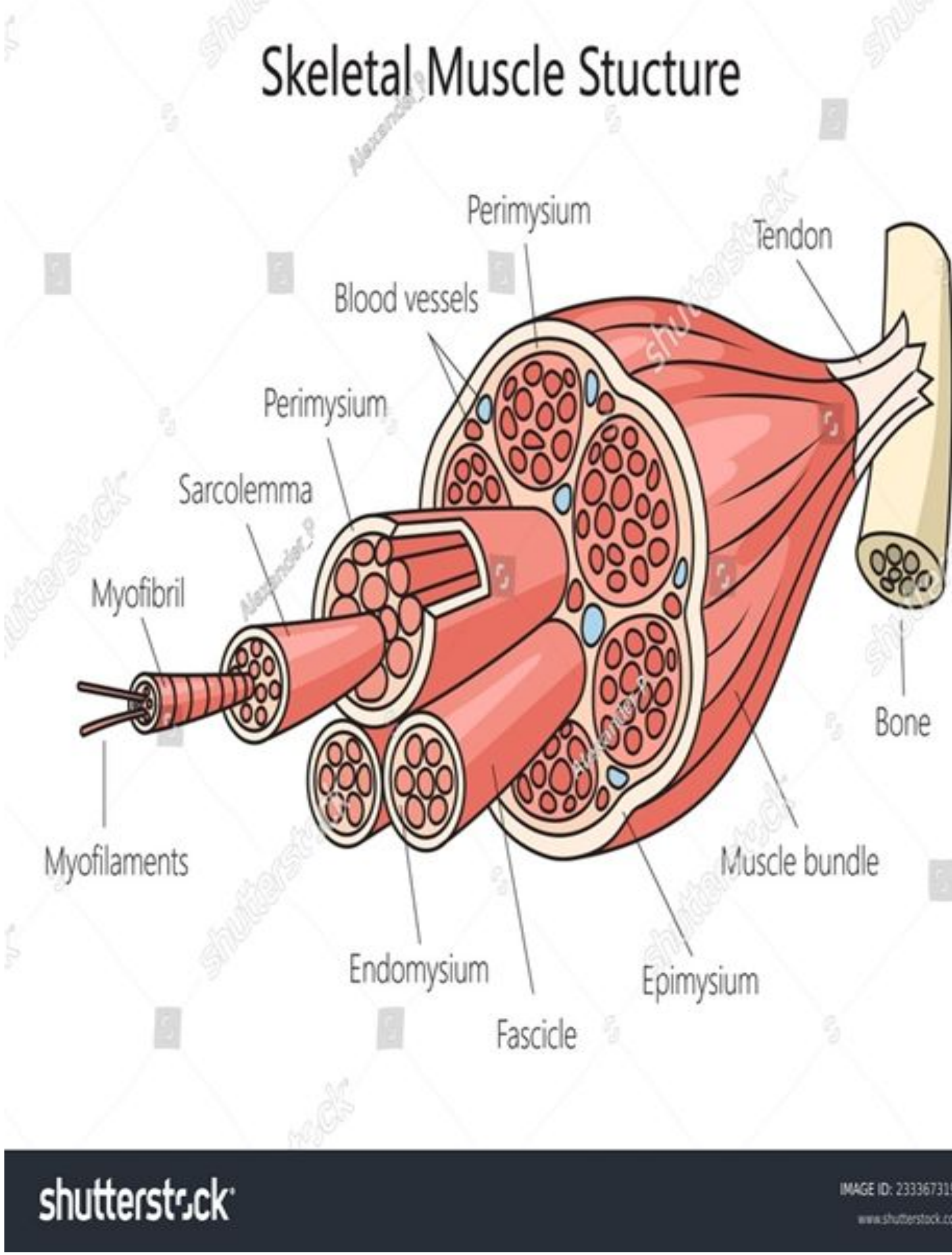
Muscle Fibers

- Skeletal muscle is made up of many cylindrical multinucleated muscle cells (fibers).
- The fibers (cell) can be 10 to 100 ten micron in diameter and can be hundreds of centimeters long.
- Each muscle cell (Fiber) is covered by a cell-membrane called **Sarcolemma.**
- Each cell contains between a few hundreds to a few thousands Myofibrils.

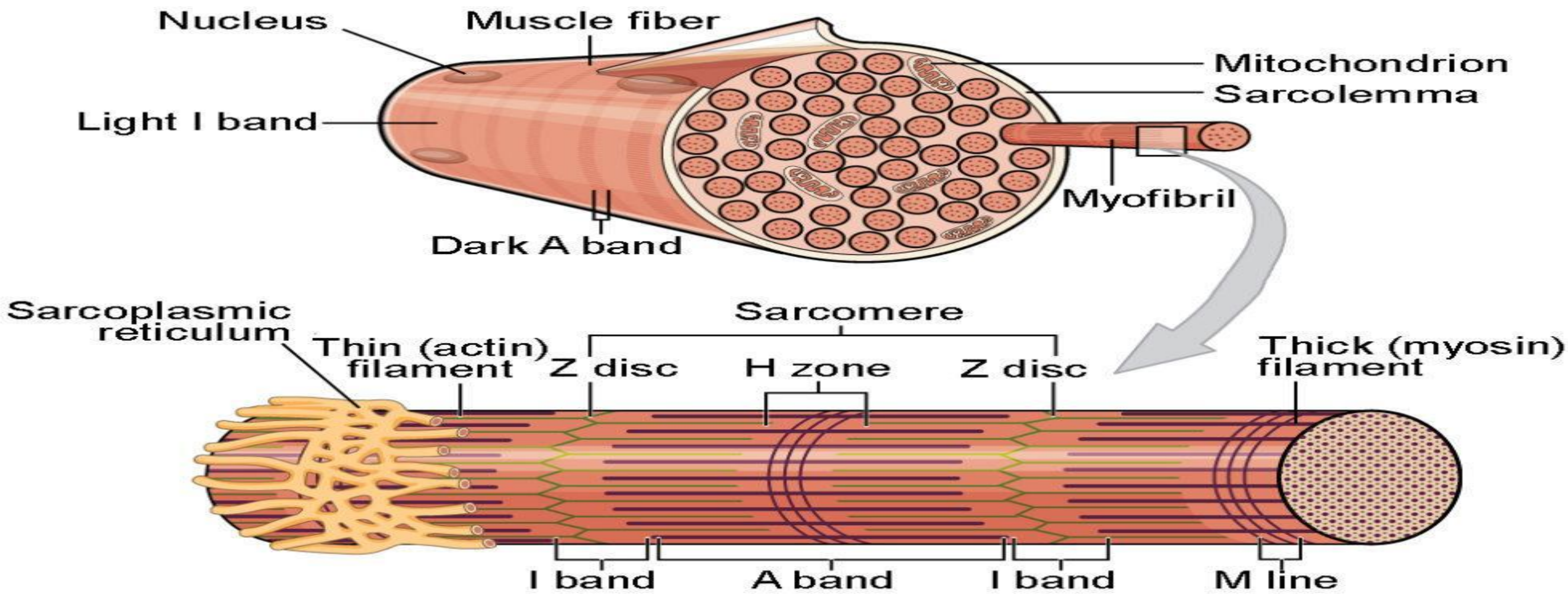
(are long filaments that run parallel to each other to form muscle (myo) fibers

Each Myofibril contains :**Actin (thin)** filaments and **Myosin (thick)** filaments .

[Each myofibril is made up of 3000 Actin and 1500 Myosin]



Each myofibril is striated and consisting of : Light and dark bands
Dark bands (A-bands),"A, band is mainly formed of thick filament (myosin)".
Light (I-bands) it has a dark line in the middle called (Z band)
"I band is formed by thin filament (actin)".
The light bands contain only actin filaments and are called I bands.



Muscle Fibers

Sarcoplasm : The term sarcoplasm refers to the cytoplasm of a muscle cell and is where metabolic processes of the cell take place. It is surrounded by a plasma membrane called the sarcolemma, which regulates the movement of various materials into or out of the muscle cell.

Sarcoplasmic reticulum : The sarcoplasmic reticulum is an intracellular structure that stores and regulates calcium in the cell.It is endoplasmic reticulum inside sarcoplasm full of Ca+.

T- tubules :. T-tubules, or transverse tubules, are deep invaginations of the sarcolemma in muscle cells, Extend from one side of muscle to other.

* What is the function of T-tubules ?

Transmission of Action potentials occur along transverse tubules (t-tubules) that penetrate all the way through the muscle fiber from one side of the fiber to the

- **Sarcomere :** The portion of a myofibril that lies between two successive Z discs is called a sarcomere
  It is the contractile unit of muscle,

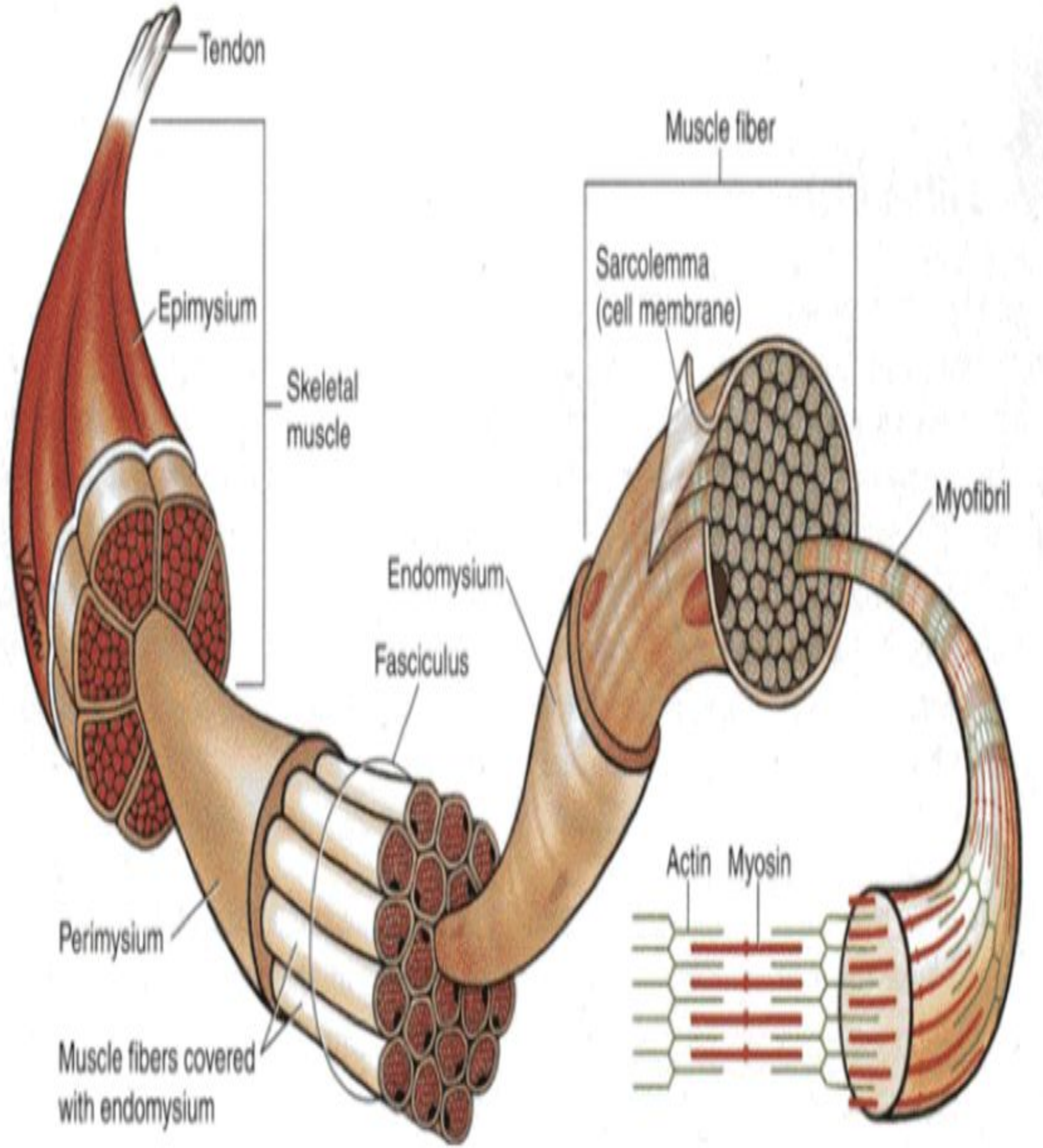
Z discs (lines) : it is the area where two actin filaments connect and transverse the I bands. It lines extend all way across myofibrils.

Muscle cell membrane Sarcolema

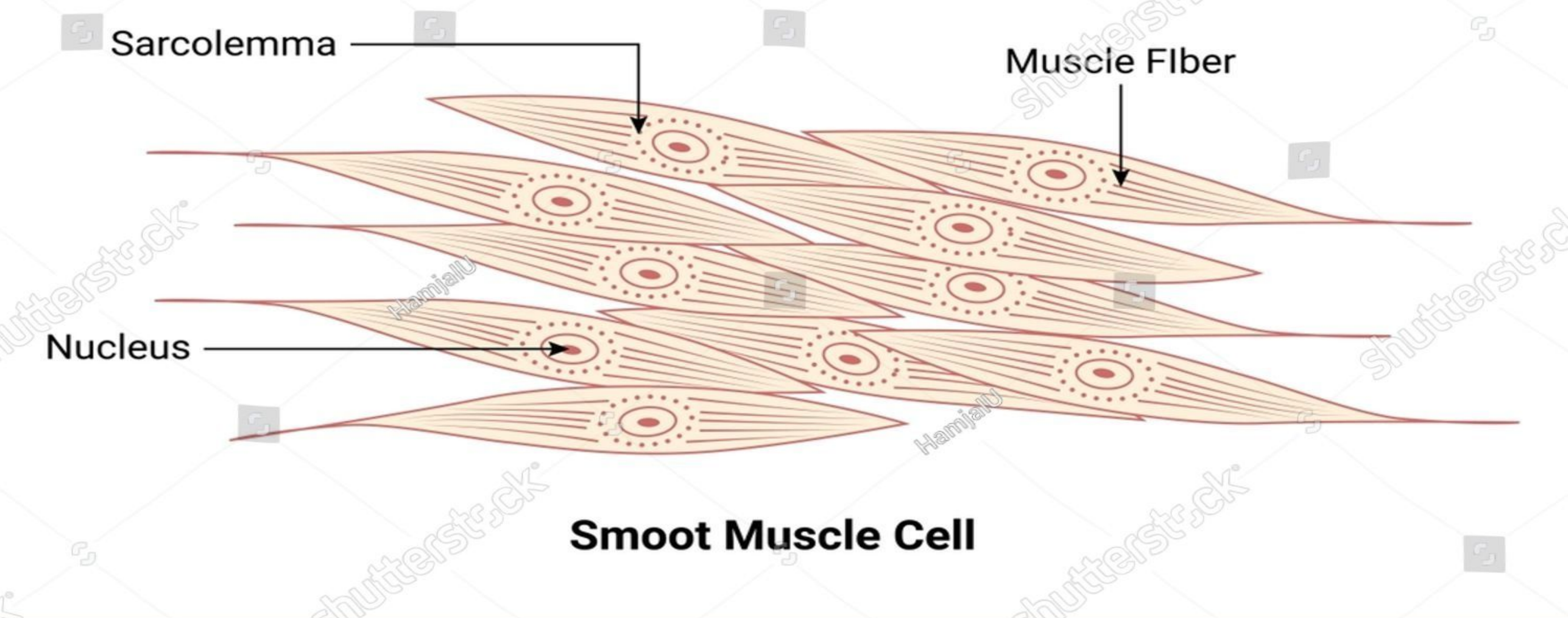
In skeletal muscle, the sarcolemma is the cell membrane that surrounds each muscle fiber, forming a barrier between the extracellular and intracellular spaces, and is crucial for transmitting electrical signals and initiating muscle contraction.

Cytoplasm of the muscle cell Sarcoplasm

The thick and thin filaments, along with their associated myofibril proteins, are responsible for muscle contraction.



SMOOTH MUSCLE



What is smooth muscle?

Smooth muscle is a type of muscle found in many places inside your body.

The smooth muscles throughout your body work without you thinking about them.

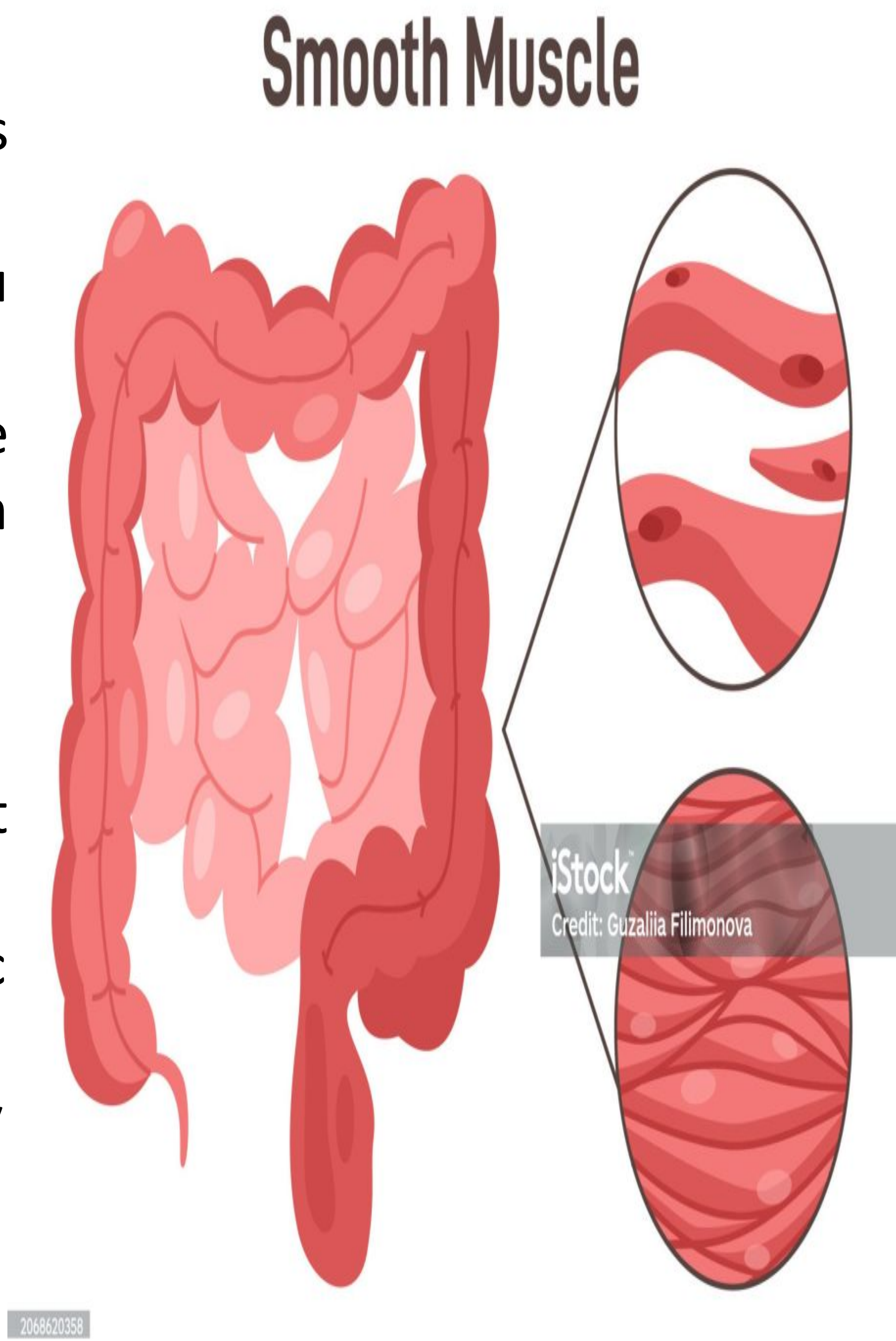
The shape of smooth muscle is fusiform, which is round in the center and tapering at each end (3-10 μm thick and 20-200 μm long).

The nucleus is located in the center and takes a cigar-like shape during contraction.

Smooth muscle contains thick and thin filaments that do not arrange into sarcomeres, resulting in a non-striated pattern

Smooth muscle can tense and relax but has greater elastic properties than striated muscle.

This quality is important in organ systems like the urinary bladder, where the preservation of contractile tone is a necessity

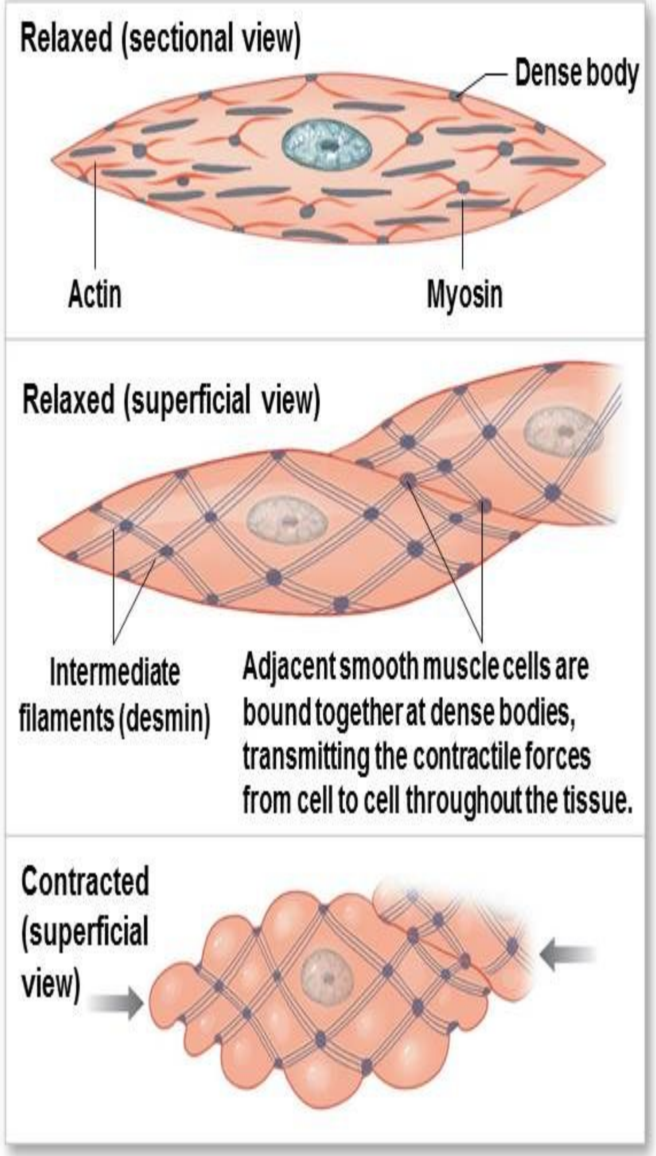


As per cross sectional area smooth muscle is as strong as skeletal muscle and smooth muscle is highly resistant to fatigue.

- ❖ The resting membrane potential of smooth muscle is usually about –50 to –60 mv

- ❖ The neurotransmitter at the neuromuscular junction of skeletal muscle is only acetylcholine ,while of smooth muscle are acetylcholine or norepinephrine

Figure 10-23b Smooth Muscle Tissue



b A single relaxed smooth muscle cell is spindle shaped and has no striations. Note the changes in cell shape as contraction occurs.

Smooth muscle is found in places throughout your body, like your:

Airways

Blood vessels

Digestive system

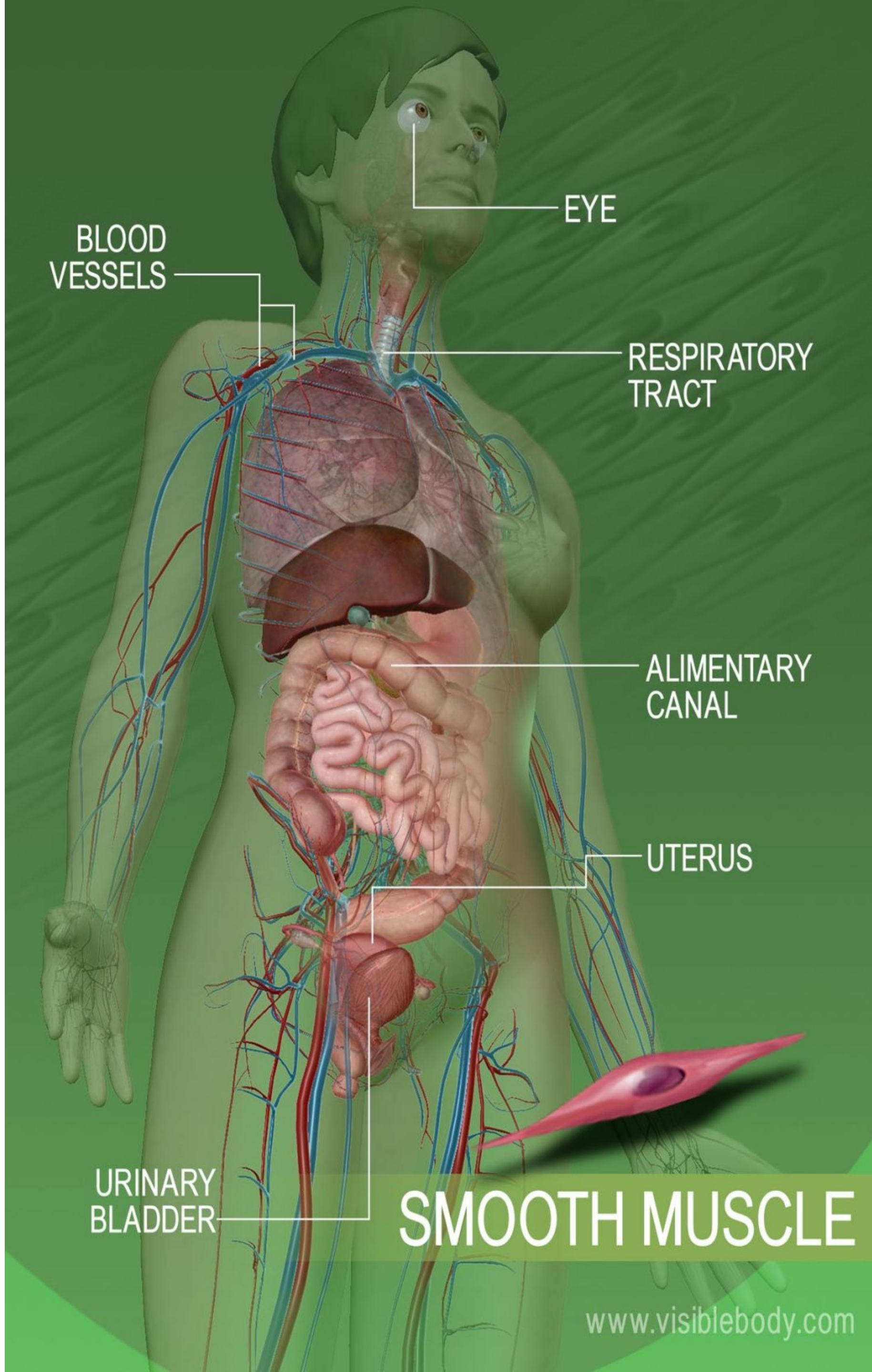
Eyes

Lymph vessels

Skin

Urinary and reproductive systems

control involuntary functions, and lack striations.



Function

- The primary function of smooth muscle is contraction.
- For simplicity, the basic functions of smooth muscle in the organ systems appear listed below.
- Gastrointestinal tract - propulsion of the food bolus
- Cardiovascular - regulation of blood flow and pressure via vascular resistance
- Renal - regulation of urine flow
- Genital - contractions during pregnancy, propulsion of sperm
- Respiratory tract - regulation of bronchiole diameter
- Integument - raises hair with erector pili muscle
- Sensory - dilation and constriction of the pupil as well as changing lens shape

Smooth muscle consists of two types: single-unit and multi-unit.

Single-unit smooth muscle

In smooth muscle, single-unit muscle cells are interconnected via gap junctions, allowing for coordinated contractions as a single unit.
The intercellular communication allows ions and molecules to diffuse between cells giving rise to calcium waves. This unique property of single-unit smooth muscle allows for synchronous contraction to occur.
Single-unit smooth muscle tissue contains gap junctions to synchronize membrane depolarization and contractions so that the muscle contracts as a single unit.
Most smooth muscles—including those in the digestive tract and uterus—are single-unit

Multi-unit smooth muscle

- Multi-unit smooth muscle differs from single-unit in that each smooth-muscle cell receives its own synaptic input, allowing for the multi-unit smooth muscle to have much finer control.

Found in:

a. Walls of large arteries
b. Airways of lungs (Bronchioles)
c. muscles attached to hair follicles
d. Radial and circular muscles of the iris of the eye

What does smooth muscle do?

Smooth muscle has a few main jobs:

Passage width control. Some of your smooth muscles manage the width of passages inside your body.

Examples of this include your blood and lymph vessels and airways.

Transporting substances. Smooth muscles can flex together to push what's inside a tube like passage. This is how multiple sections of your digestive tract work.

Gatekeeping. Most muscles stay relaxed and only flex when needed. The smooth muscle in sphincters, like in your bladder (anus), stays flexed and only relaxes when needed.

Smooth muscles in certain places can also have very specific jobs. Some examples include:

Skin. Smooth muscle is what causes hairs to stand on end, like with goose bumps.

Eyes. Smooth muscles in the ciliary body of your eye pull on fibers called zonules to control how your eyes focus. Smooth muscles in your iris also control how your pupils dilate or constrict.

Uterus Smooth muscles are where contractions happen during labor

What are the common conditions and disorders that affect smooth muscle?

Smooth muscle is vulnerable to many of the same conditions that can affect any muscle, especially injuries or paralysis . Some other conditions that can affect smooth muscle include:

- **Anti-smooth muscle autoantibodies**. This is when your immune system turns on and damages your smooth muscle. This is more likely to happen in people with autoimmune hepatitis and similar conditions.
- **Muscular dystrophies**. These are conditions that gradually make your muscles stop working properly.
- **Spasms**. Smooth muscle can twitch or contract uncontrollably. Some specific examples include spasms in your blood vessels (vasospasm) or airways (asthma).
- **Visceral myopathies**. These are a group of smooth muscle diseases that interfere with how these muscles work. They can be deadly when they affect smooth muscle in places like your bladder or large intestine. They're often genetic.

Smooth muscle dysfunction can manifest in various diseases, including
hypertension, asthma,
gastrointestinal motility issues,
and certain forms of muscular dystrophy.
These conditions can affect blood vessels, airways, the digestive tract, and other systems where smooth muscle plays a role.

The most common symptoms of a muscle injury or health condition include:
Muscle pain.
Muscle weakness.
Stiffness.
Muscle spasms (cramps).
Swelling.
Bruising.
Skin discoloration.

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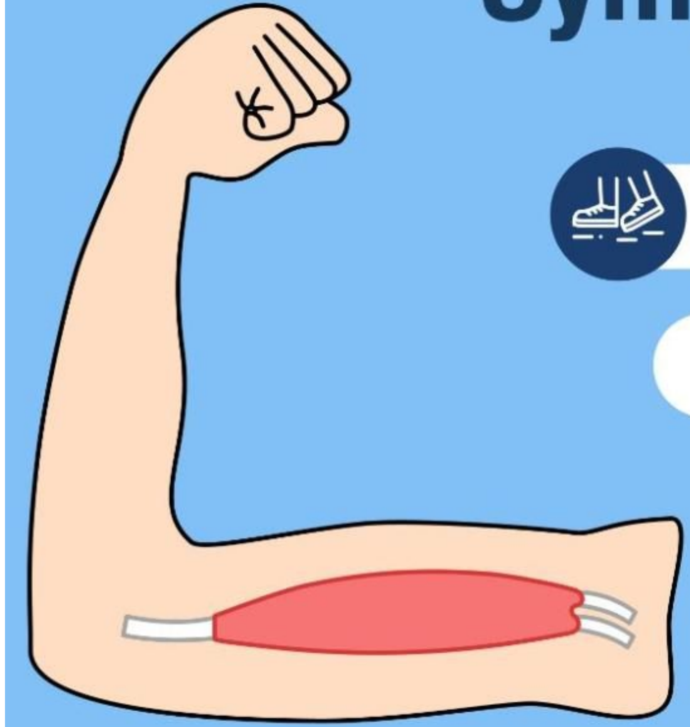
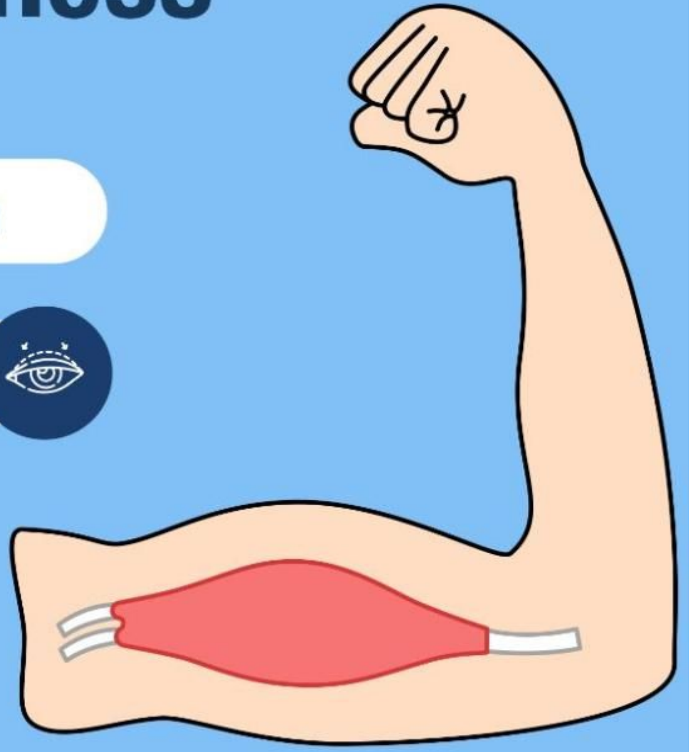
LIFE'S ON

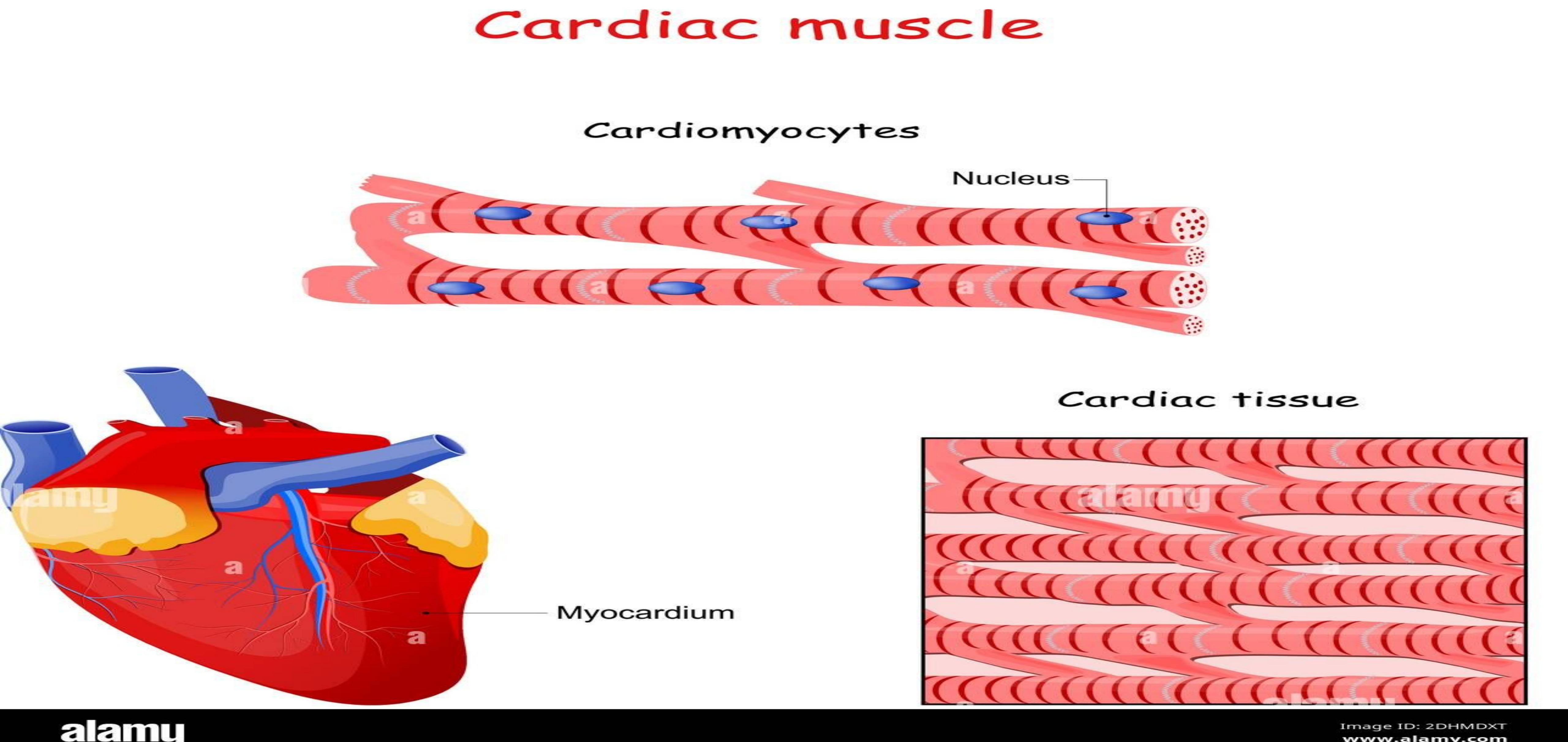
Symptoms of Muscle Weakness



- Difficulty moving
- Slurred speech
- Fatigue
- Drooping eyelids
- Breathing difficulties
- Struggling to do daily tasks
- Difficulty swallowing



CARDIAC MUSCLE



Cardiac muscle also called the myocardium, is one of three major categories of muscles found within the human body, along with smooth muscle and skeletal muscle.

Cardiac muscle, like skeletal muscle, is made up of sarcomeres that allow for contractility. However, unlike skeletal muscle, cardiac muscle is under involuntary control.

The cardiac muscle is responsible for the contractility of the heart.

The cardiac muscle must contract with enough force and enough blood to supply the metabolic demands of the entire body.

This concept is termed cardiac output and is defined as heart rate x stroke volume,

The duration of cardiac action potential is much longer(calcium influx prolongs the duration of the action potential and produces a characteristic plateau phase) than in nerve or skeletal muscle fiber. In a typical nerve, the action potential duration is about 1 ms.

In skeletal muscle cells, the action potential duration is approximately 2-5 ms. In contrast, the duration of cardiac action potentials range from 200 to 400 ms.

Structure of cardiac muscle

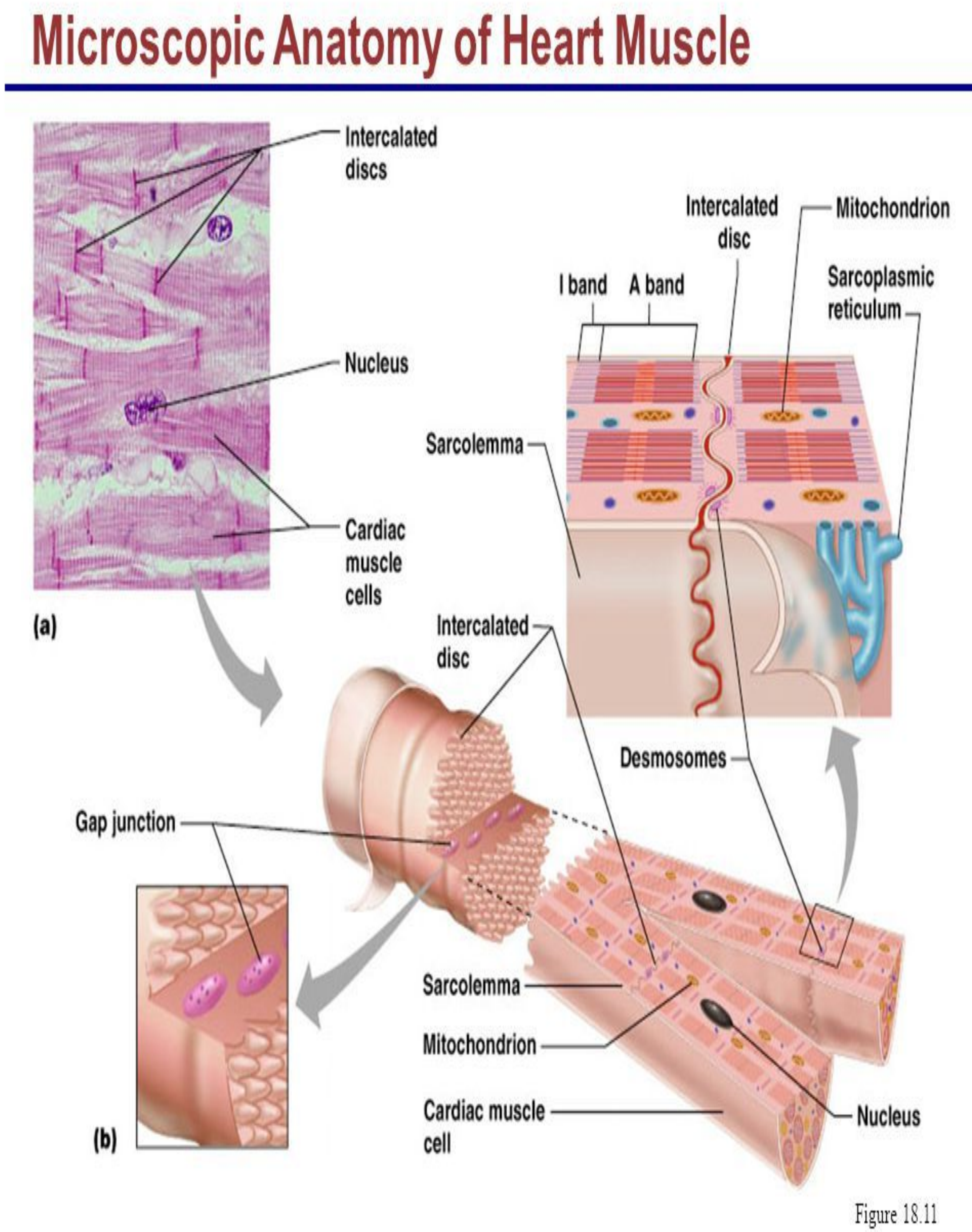
Cardiac muscle cells (cardiomyocytes) are short branched striated muscle cells

which, contain many mitochondria, and are under involuntary control.

Each myocyte contains a single, centrally located nucleus surrounded by a cell membrane known as the sarcolemma.

The sarcolemma of cardiac muscle cells contains voltage-gated calcium channels, specialized ion channels that skeletal muscle does not possess.

Cardiac muscle cell are Connected with gap junctions which transmit electrical activity between cells So, cardiac myocytes act as a single functional unit (syncitium)



Properties of cardiac muscle

1.Rhythmicity

2. Excitability

3. Conductivity

4. Contractility

Rhythmicity

Rhythmicity means the ability of the heart to beat regularly without external stimulation.

It is myogenic in origin not neurogenic

The nodal fibers and conducting system are self excitable.

The cells of SAN; (posterior wall of right atrium) is the primary pacemaker of the heart

Conductivity

The ability to conduct impulse from one cell to another facilitated by the presence of gap junctions that transmit electrical currents

From SAN→ atrial muscle & atrioventricular node (AVN)
From AVN (slowest) → atrioventricular (AV) bundle (bundle of His) →left & right bundles →purkinje fibres (fastest)

Excitability

The heart muscle responds to stimuli which may be mechanical, electrical or chemical

Refractory Period

The heart muscle refractory period refers to a period of time in which the heart muscle is unable to be stimulated again, preventing double counting of events and allowing for the resetting of the heart's natural rhythm.

The refractory period of the myocardial fibers is of much longer duration than that of skeletal muscle fibers

Nerve supply of the Heart

A) Sympathetic supply:

1.↑es all cardiac properties

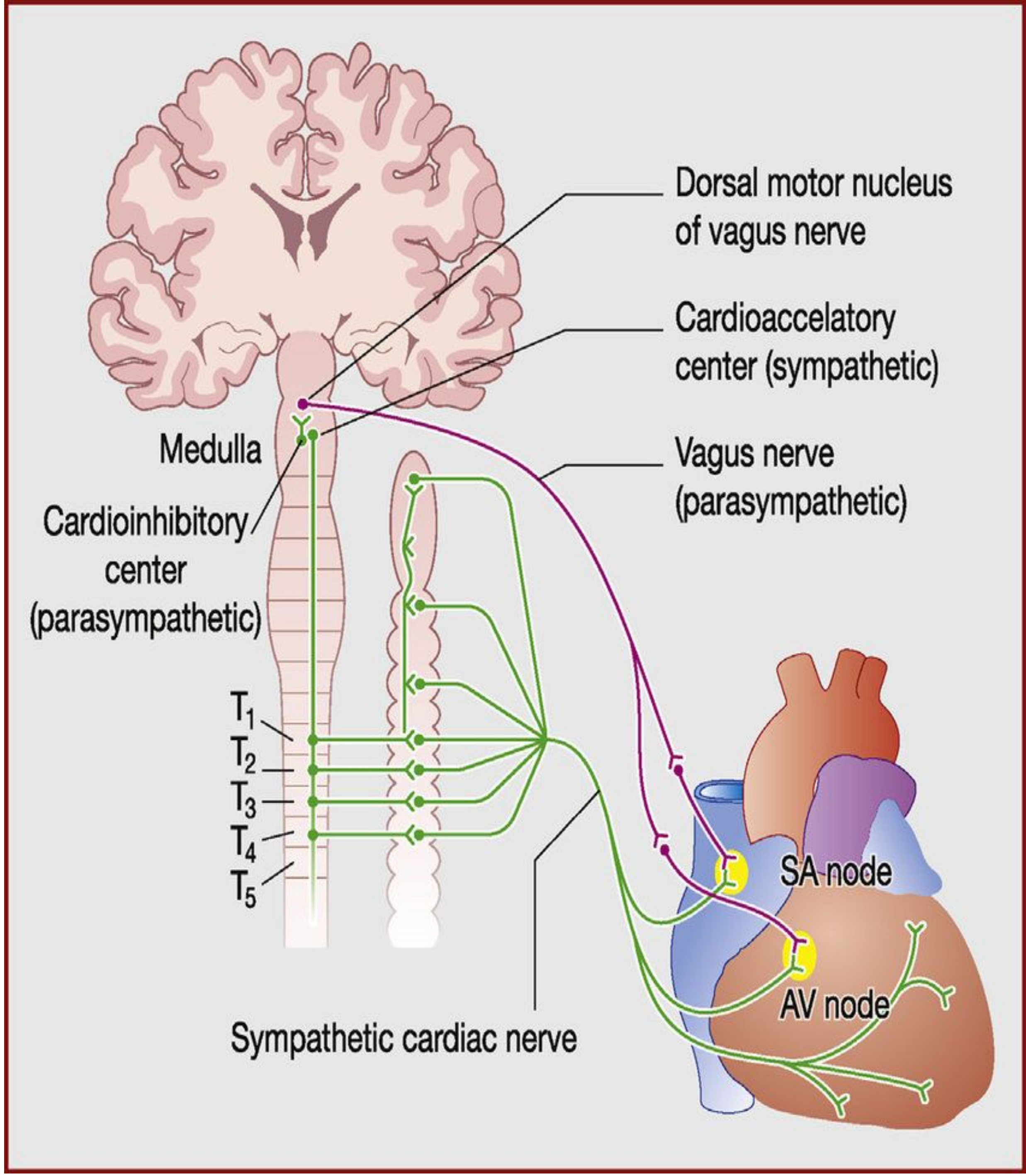
2. ↑es the coronary blood flow.

B) Parasympathetic supply:

1.↓es all cardiac properties except the ventricles

(not supplied by vagus nerve)

2.↓es the coronary blood flow

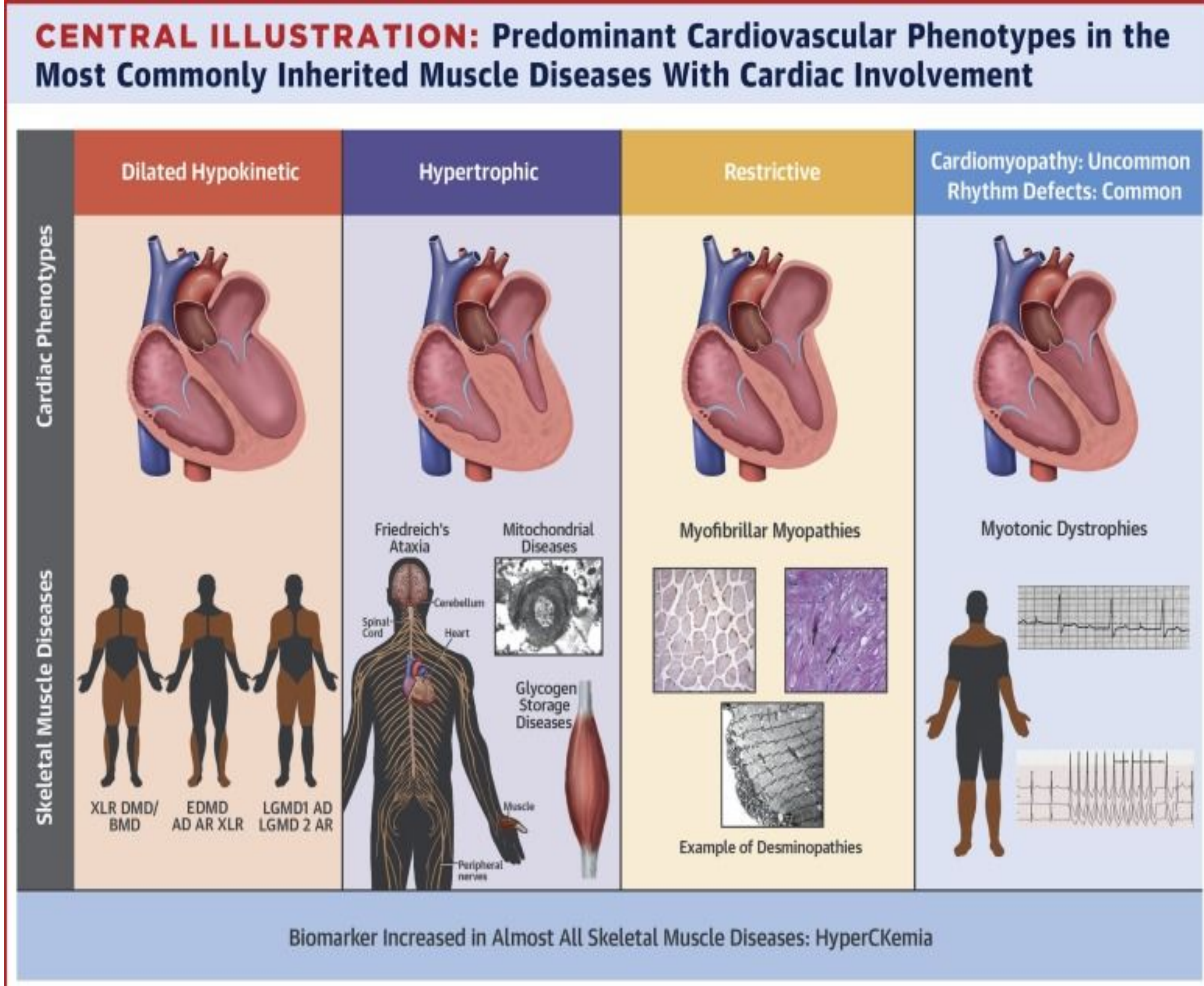


Cardiac Muscle Disorders

Dilated Cardiomyopathy

Hypertrophic Cardiomyopathy

Restrictive



Arbustini, E. et al. J Am Coll Cardiol. 2018;72(20):2485-506.

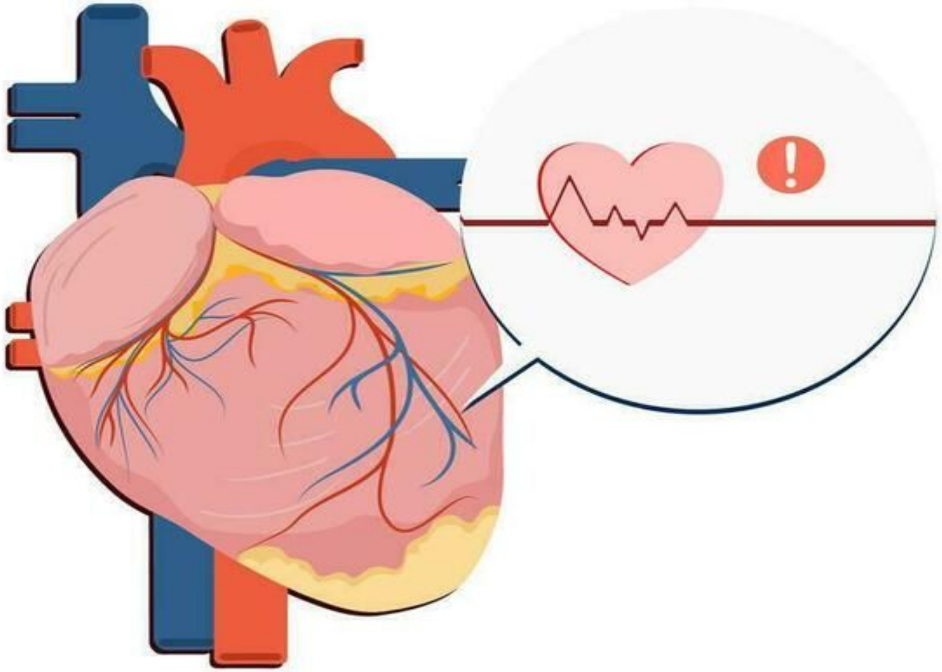
Warning Signs of Heart Disease

EVERYDAY HEALTH

Symptoms of

Heart Disease

Heart Failure



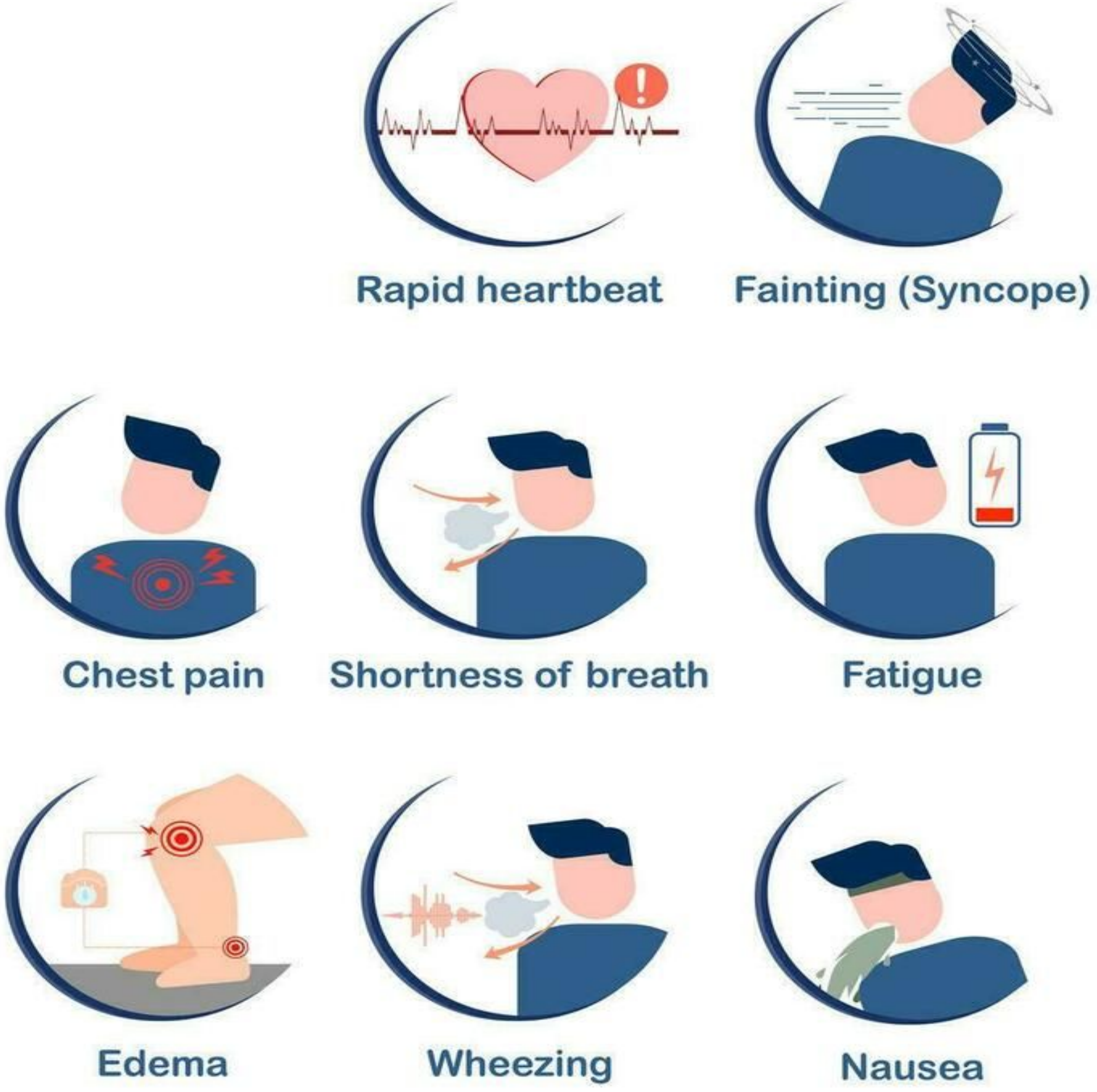


Table 4.2: summary of the differences among the three types of muscles

	Skeletal muscles	Cardiac muscles	Smooth muscles
Microscopic appearance	Striated; actin and myosin arranged in sarcomeres	Striated; actin and myosin arranged in sarcomeres	Not striated; no sarcomeres, actin anchored into dense bodies and cell membrane
Innervation	Somatic nervous system (alpha motor neurons)	Autonomic nervous system	Autonomic nervous system
Level of Control	Under voluntary control	Under involuntary Control	Under involuntary control
Presence of Troponin	Yes	Yes	No
Presence of T Tubules	Yes	Yes	No
Source of Increased Cytosolic Ca^{2+}	Sarcoplasmic reticulum	From ECF and from sarcoplasmic reticulum	From ECF and from sarcoplasmic reticulum
Site of Ca^{2+} regulation	Troponin in thin filaments	Troponin in thin filaments	Myosin in thick filaments
Presence of Gap Junctions	No	Yes	Yes in single-unit smooth muscle
Length–Tension Relationship	Yes but limited	Yes	No


thank you

Skeletal Muscles

Skeletal muscles are muscles which are attached to the skeleton.

A muscle consists of packages of muscle cells called **fascicles**

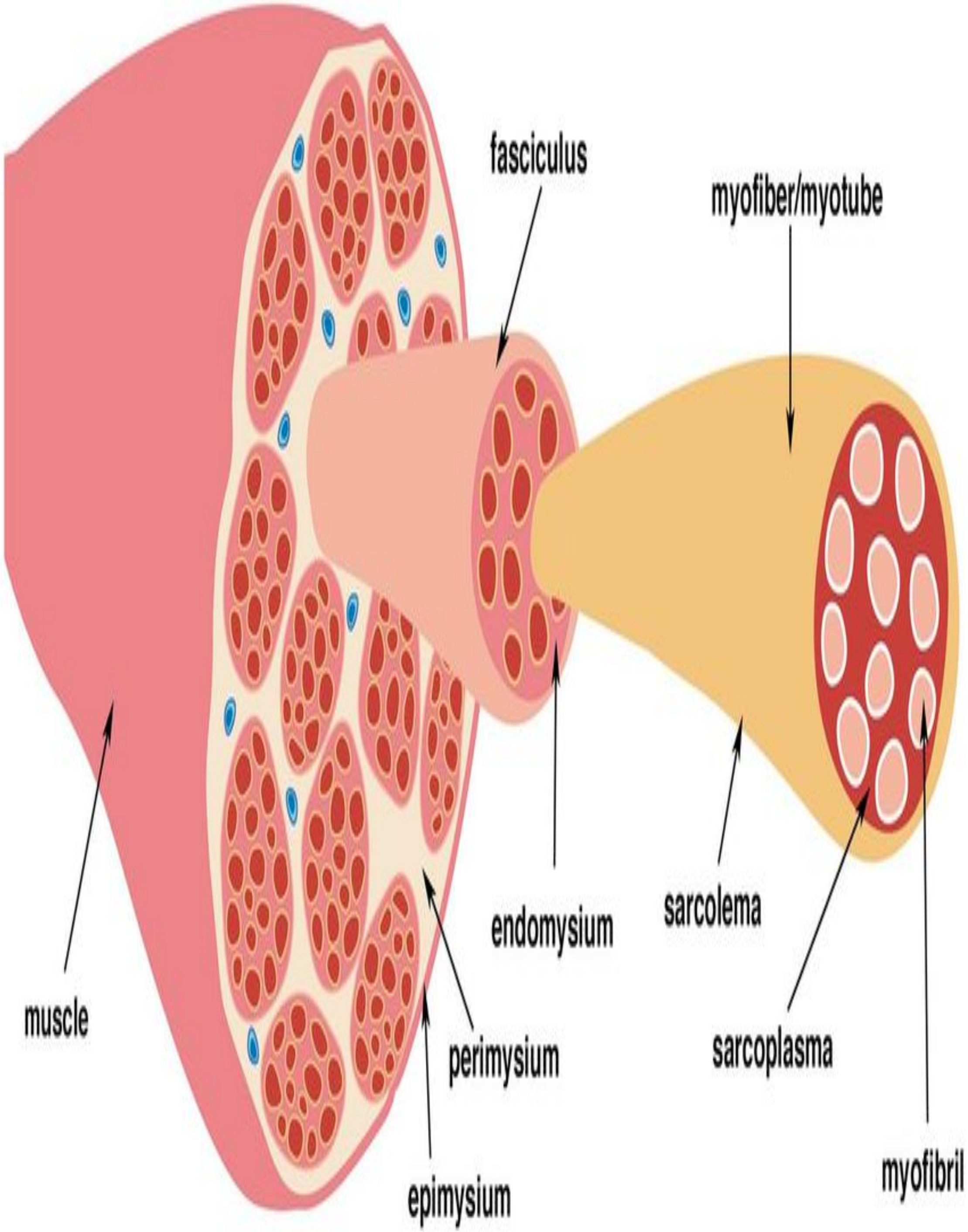
A muscle cell is long and spindle shaped

These muscles are cylindrical and are characterized by the presence of multiple nuclei and striations.

Striations are caused by a repeating unit in the muscle fiber called the sarcomere.

Sarcomeres have alternating thick filaments of myosin that give a dark band (red) and a thin filament of actin that gives a light band (white).

This alternating dark and light band gives rise to the striated appearance of skeletal muscles.



- Structure

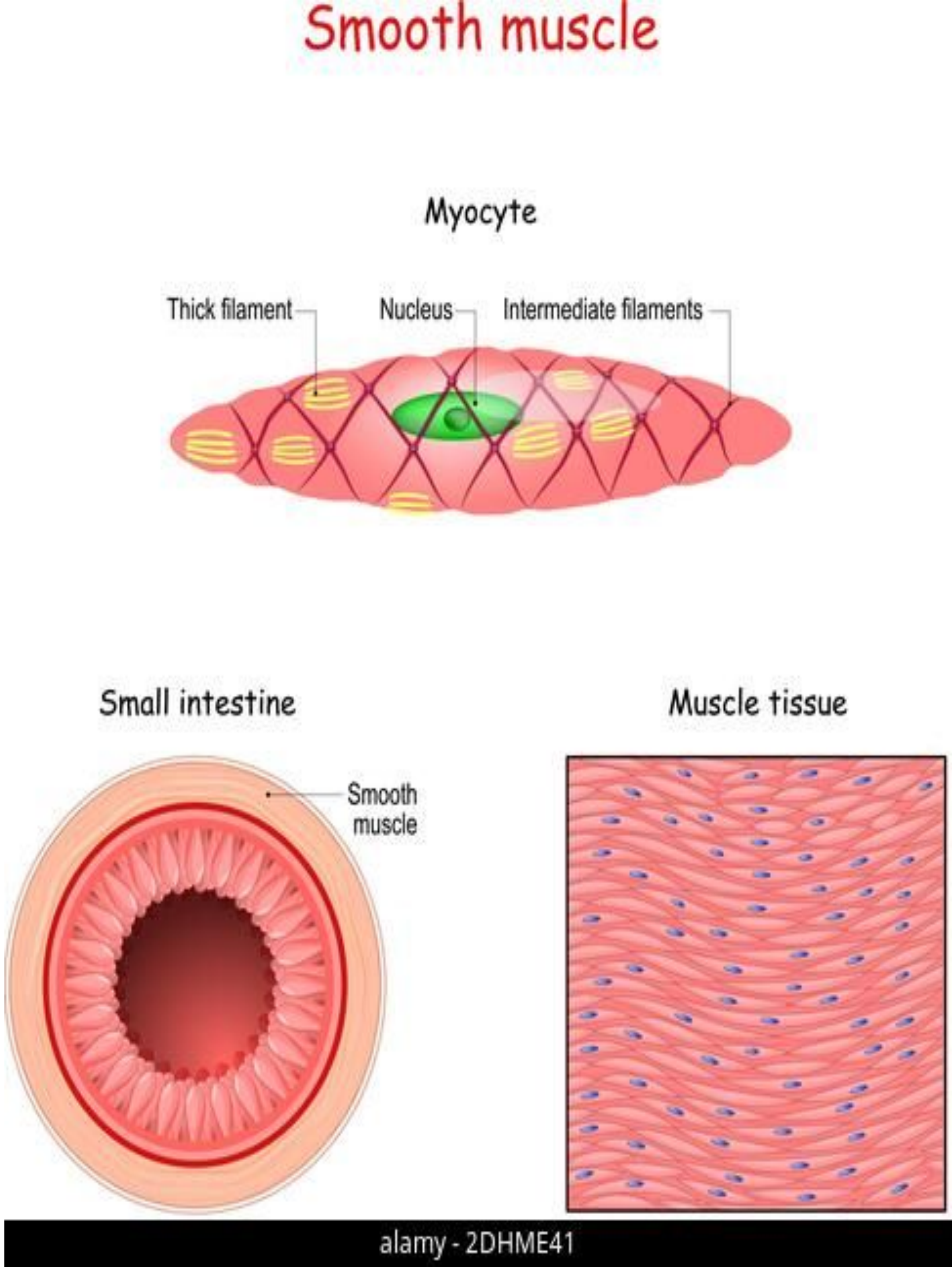
The shape of smooth muscle is fusiform, which is round in the center and tapering at each end (3-10 µm thick and 20-200 µm long).

The nucleus is located in the center and takes a cigar-like shape during contraction.

Smooth muscle contains thick and thin filaments that do not arrange into sarcomeres, resulting in a non-striated pattern

Smooth muscle cytoplasm contains large amounts of actin and myosin. Actin and myosin act as the main proteins involved in muscle contraction.

The cytoplasm consists mainly of myo filaments.



Muscle striations are not visible in smooth muscle, so the sarcomere relationship of myosin to actin does not exists in smooth muscle. However, per cross sectional area smooth muscle is as strong as skeletal muscle and smooth muscle is highly resistant to fatigue.

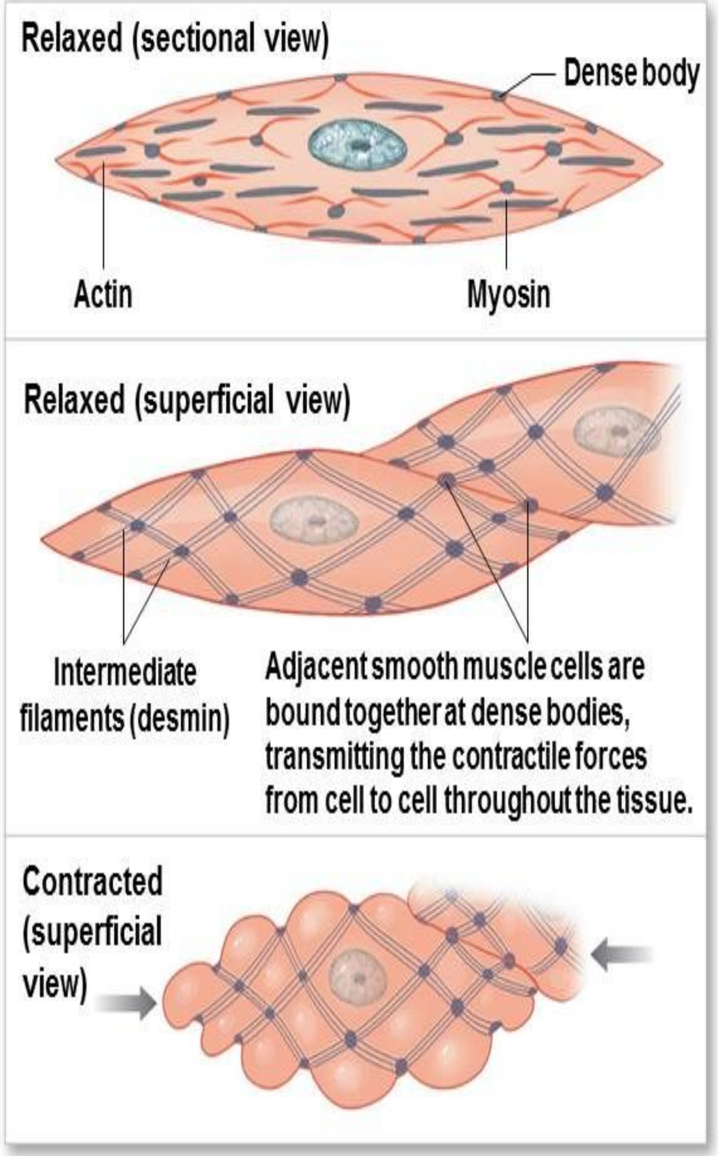
❖ The resting membrane potential of smooth muscle is usually about –50 to –60 mv

The junction between the nerve terminals and smooth muscles are called diffuse junction

❖ The neurotransmitter at the neuromuscular junction of skeletal muscle is only acetylcholine,

while of smooth muscle are acetylcholine or norepinephrine

Figure 10-23b Smooth Muscle Tissue



b A single relaxed smooth muscle cell is spindle shaped and has no striations. Note the changes in cell shape as contraction occurs.

Skeletal Muscles

- Skeletal muscles are also known as striated or <u>voluntary</u> muscles.
- They form a significant part of the musculoskeletal system and are responsible for the locomotion and movement of body parts.
- These muscles are cylindrical and are characterized by the presence of multiple nuclei and striations.
- Striations are caused by a repeating unit in the muscle fiber called the sarcomere.
- Sarcomeres have alternating thick filaments of myosin that give a dark band (red) and a thin filament of actin that gives a light band (white). This alternating dark and light band gives rise to the striated appearance of skeletal muscles.
- Skeletal muscles are attached to the bones by tendons. These muscles are under conscious control and are responsible for voluntary movements such as walking, running, and lifting. They are vital in providing structural support and maintaining posture.

Gap junctions between adjacent cardiomyocytes allow for the propagation of coordinated action potentials from one cell to the next in a phenomenon known as electrical coupling

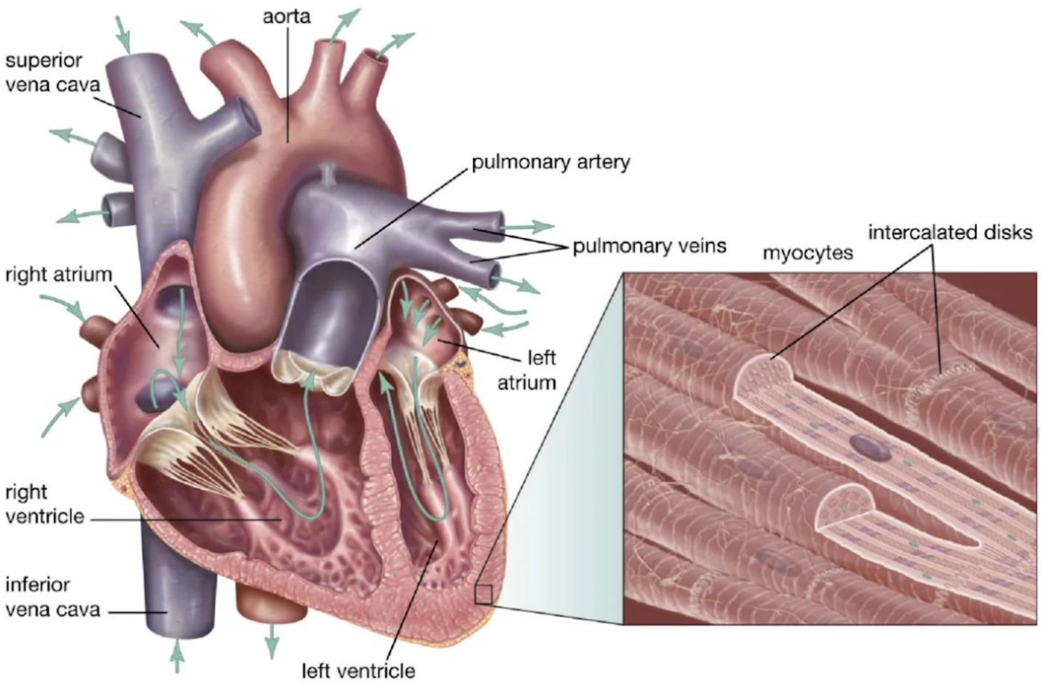
The functional unit of cardiomyocyte contraction is the sarcomere, which consists of thick (myosin) and thin (actin) filaments,

The sarcolemma is the cardiomyocyte plasma membrane containing transverse tubules (t-tubules).

T-tubules regulate the cardiac ECC by concentrating voltage-gated

Understanding the Key Properties of Cardiac Muscle

- Cardiac muscle's automatic rhythmic contractions
- Excitability for rapid response
- Efficient electrical signal conduction
- Forceful, regulated contractions
- Elasticity for structural integrity
- Continuous, fatigue-resistant function



Smooth muscle consists of two types: single-unit and multi-unit.
Single-unit smooth muscle
In smooth muscle, single-unit muscle cells are interconnected via gap junctions, allowing for coordinated contractions as a single unit.
It consists of multiple cells connected through connexins (Connexins are trans membrane proteins that assemble to form gap junctions, which allow intercellular movement of molecules and ions)that can become stimulated in a synchronous pattern from only one synaptic input.
Connexins allow for cell-to-cell communication between groups of single-unit smooth muscle cells. This intercellular communication allows ions and molecules to diffuse between cells giving rise to calcium waves. This unique property of single-unit smooth muscle allows for synchronous contraction to occur.
Most smooth muscles—including those in the digestive tract and uterus—are single-unit.
Single-unit smooth muscles have numerous gap junctions (electrical synapses) between adjacent cells that weld them together electrically; they thus behave as a single unit, much like cardiac muscle.
Single-unit smooth muscle consists of multiple cells connected through connexins that can become stimulated in a synchronous pattern from only one synaptic input.
Single-unit smooth muscle tissue contains gap junctions to synchronize membrane

SINGLE UNIT SMOOTH MUSCLE	MULTIUNIT SMOOTH MUSCLE
These are connected using gap junctions	Gap junctions are not seen
They act as a single functional unit	They act as multiple individual units
Syncytium is present	Syncytium is absent
They show the activity of the pacemaker	The activity of the pacemaker is not observed
Affected by muscle stretch	No affect by muscle stretch
contraction is not localized, rather found as a single unit.	6. contraction is discrete and localized
The autonomic nervous system is said to modify the response caused by these muscles	7. The autonomic nervous system can not modify the response caused by these muscles.
These are found in Gut, uterus, and bladder	8. These are found in eye muscles, in and large blood vessels

Rules controlling Contractility

1-All or None Law

The cardiac muscle contracts either maximally or not at all (under constant conditions)

The Atria contract as one unit & the ventricles contract as one unit

This is significant for efficient pumping of the blood

2-Staircase or Treppe Phenomenon

Rapidly Repeated stimulation of the cardiac muscle produce gradual increase in the strength of contraction

The earlier contractions produce better conditions (heat, less viscosity between muscle fiber, more Ca) for the following contraction

3-Starling Law

Within limits, the greater the initial length of cardiac muscle fibre (stretch), the greater the force of contraction

The initial length is determined by the volume of blood filling ventricles at end of diastole (end-diastolic volume; EDV)